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EDA dashboard

ML Models



EDA Dashboard

Air Quality · 13 Tabs

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Tab 1 — Data Overview

Records 9,357 Features 23 GT Sensors 4 Period Mar 20... Freq Hourly

First 10 Rows

	Date	Time	CO(GT)	PT08.S1(CO)	C6H6(GT)
0	3/10/2004	18:00:00	2.6	1360	11.9
1	3/10/2004	19:00:00	2	1292	9.4
2	3/10/2004	20:00:00	2.2	1402	9
3	3/10/2004	21:00:00	2.2	1376	9.2
4	3/10/2004	22:00:00	1.6	1272	6.5
5	3/10/2004	23:00:00	1.2	1197	4.7
6	3/11/2004	0:00:00	1.2	1185	3.6
7	3/11/2004	1:00:00	1	1136	3.3
8	3/11/2004	2:00:00	0.9	1094	2.3
9	3/11/2004	3:00:00	0.6	1010	1.7

Column Types & Nulls

	Column	Dtype	Nulls	Null %
13	AH	float64	0	0
14	Datetime	datetime64[	0	0
15	Hour	int32	0	0
16	Day	int32	0	0
17	Month	int32	0	0
18	DayOfWeek	int32	0	0
19	Weekend	int32	0	0
20	Season	object	0	0
21	is_rush_hour	int32	0	0
22	high_pollution	int32	0	0

Descriptive Statistics — Pollutants

	CO(GT)	C6H6(GT)	NOx(GT)	NO2(GT)	T	RH	AH
count	9357	9357	9357	9357	9357	9357	9357
mean	2.089	10.009	235.179	112.373	18.298	49.249	
std	1.323	7.312	195.091	43.949	8.658	16.975	
min	0.1	0.1	2	2	-1.9	9.2	
25%	1.2	4.6	112	86	12	36.6	
50%	1.8	8.2	180	109	17.8	49.6	
75%	2.6	13.6	284	133	24.1	61.9	
max	11.9	63.7	1479	340	44.6	88.7	

****Dataset:**** UCI Air Quality — Multisensor device deployed in a polluted Italian city road, recording hourly readings from March 2004 to April 2005.

****200 marker:**** Sensor null code — replaced with NaN before any analysis.

****NMHC(GT)**** dropped — 90.3% missing, not usable for analysis.

****CO(GT), NOx(GT), NO2(GT)**** had ~19% missing — imputed with column median.

Column Dictionary

	Column	Type	Description
0	CO(GT)	GT	Carbon Monoxide — reference analyzer (mg/m ³)
1	PT08.S1(CO)	Sensor	Metal oxide sensor response (correlated with CO)
2	C6H6(GT)	GT	Benzene — reference analyzer (µg/m ³)
3	PT08.S2(NMHC)	Sensor	Metal oxide sensor (correlated with NMHC)
4	NOx(GT)	GT	Nitrogen Oxides — reference analyzer (µg/m ³)
5	PT08.S3(NOx)	Sensor	Metal oxide sensor (correlated with NOx)
6	NO2(GT)	GT	Nitrogen Dioxide — reference analyzer (µg/m ³)
7	PT08.S4(NO2)	Sensor	Metal oxide sensor (correlated with NO2)
8	PT08.S5(O3)	Sensor	Metal oxide sensor (correlated with O3)
9	T	Weather	Temperature (°C)